Mathematics - Graduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If f(2 x) = x + 5 and f(g(6)) = 13, then 2 * g(6) =

   (a) 6
   (b) 32
   (c) 30
   (d) 12
   (e) 36

2. A shopping mall wants to conduct a survey of the people who shop at the mall. Which would give them the most representative sample?

   (a) conducting the survey at one shoe store
   (b) conducting the survey online targeting only the mall's registered customers
   (c) conducting the survey at all clothing stores
   (d) conducting the survey only on weekends
   (e) conducting the survey at the entrance to the mall

3. A function f(x) is given by f(0)=3, f(2)=7, f(4)=11, f(6)=9, f(8)=3. Approximate the area under the curve y=f(x) between x=0 and x=8 using Trapezoidal rule with n=4 subintervals.

   (a) 85.0
   (b) 60.0
   (c) 65.0
   (d) 45.0
   (e) 55.0

4. The rainfall for the last three months of school was 8.1 inches, 4.2 inches, and 0.33 inch. Estimate the total rainfall for the three months by rounding to the nearest whole number.

   (a) 13 in.
   (b) 10 in.
   (c) 11 in.
   (d) 45.0 in.
   (e) 15 in.

5. Find the area bounded by the curves $y = \cos x$ and $y = \cos^2 x$ between $x = 0$ and $x = \pi$.

   (a) 3.5
   (b) 1
   (c) 0.25
   (d) 0.5
   (e) 2.5

True / False (10 marks)

Answer True or False

6. True or False: The probability that a randomly chosen real number x in [0, 3] is less than a randomly chosen real number y in [0, 4] is 4/7.

7. True or False: The limit of the given integral expression as r approaches infinity is approximately 0.54321.

8. True or False: The value of k in the quartic equation x^4 - 18 x^3 + kx^2 + 200 x - 1984 = 0 is 86, given that the product of two roots is -32.

9. True or False: The number of ways to choose 4 balls with at least 2 red from the box is 12.

10. True or False: The smallest positive integer with factors of 16, 15, and 12 is 120.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Derek's phone number, 336 - 7624, has the property that the three-digit prefix, 336, equals the product of the last four digits, $7 \times 6 \times 2 \times 4$. How many seven-digit phone numbers beginning with 336 have this property?

12. In rectangle $ABCD$, P is a point on BC so that $\angle APD = 90^\circ$. TS is perpendicular to BC with $BP = PT$, as shown. PD intersects TS at Q. Point R is on CD such that RA passes through Q. In $\triangle PQA$, $PA = 20$, $AQ = 25$ and $QP = 15$. [asy] size(7 cm);defaultpen(fontsize(9)); real sd = 7/9 * 12; path extend(pair a, pair b) {return a–(10 * (b - a));} // Rectangle pair a = (0, 0); pair b = (0, 16); pair d = (24 + sd, 0); pair c = (d.x, b.y); draw(a–b–c–d–cycle); label("A", a, SW);label("B", b, NW);label("C", c, NE);label("D", d, SE); // Extra points and lines pair q = (24, 7); pair s = (q.x, 0); pair t = (q.x, b.y); pair r = IP(c–d, extend(a, q)); pair p = (12, b.y); draw(q–a–p–d–r–cycle);draw(t–s); label("R", r, E); label("P", p, N);label("Q", q, 1.2 * NE + 0.2 * N);label("S", s, S); label("T", t, N);

// Right angles and tick marks markscalefactor = 0.1; draw(rightanglemark(a, b, p)); draw(rightanglemark(p, t, s)); draw(rightanglemark(q, s, d));draw(rightanglemark(a, p, q)); add(pathticks(b–p, 2, spacing=3.4, s=10));add(pathticks(p–t, 2, spacing=3.5, s=10)); // Number labels label("16", midpoint(a–b), W); label("20", midpoint(a–p), NW); label("15", midpoint(p–q), NE); label("25", midpoint(a–q), 0.8 * S + E); [/asy] Find the lengths of BP and QT. When writing your answer, first write the length of BP, then a comma, and then the length of QT. For example, if you find that these lengths are 5 and 3/4, respectively, your final answer should be written "5,3/4" (without the quotes).

End of Paper